

ULX4M-LS v004

2025-07-14

Intergalaktik d.o.o.
License: CERN-OHL-S v2
Year: 2025

click on mouse pointer arrow on top of right toolbar
and double-click on sheet to open

power BANK POWER 3V3 2V5 1V1/1V2 1V35(DNP) File: power.kicad_sch	gpio BANK0 BANK7 CM4 connector File: gpio.kicad_sch
usb BANK6 CLK25 USB FPGA File: usb.kicad_sch	gpd BANK1 SW DSI File: gpd.kicad_sch
blinky 3X BTN 8X LED File: blinky.kicad_sch	ethernet ETHERNET (LAN8720A) File: ethernet.kicad_sch
ram BANK2 BANK3 File: ram.kicad_sch	SDRAM SDRAM (B428163208) File: ddr3.kicad_sch
serdes SERDES File: serdes.kicad_sch	flash BANK8 FLASH (IS25LP128F) File: flash.kicad_sch
MechanicalDrawings File: MechanicalDrawings.kicad_sch	

Page

.....

CoverPage
PowerPage
GPIO
USB
GPD
Blinky
Ethernet
SDRAM FPGA
SDRAM
SerDes
Flash
MechanicalDrawing

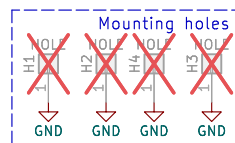
.....

Index

.....

1
2
3
4
5
6
7
8
9
10
11
12

.....



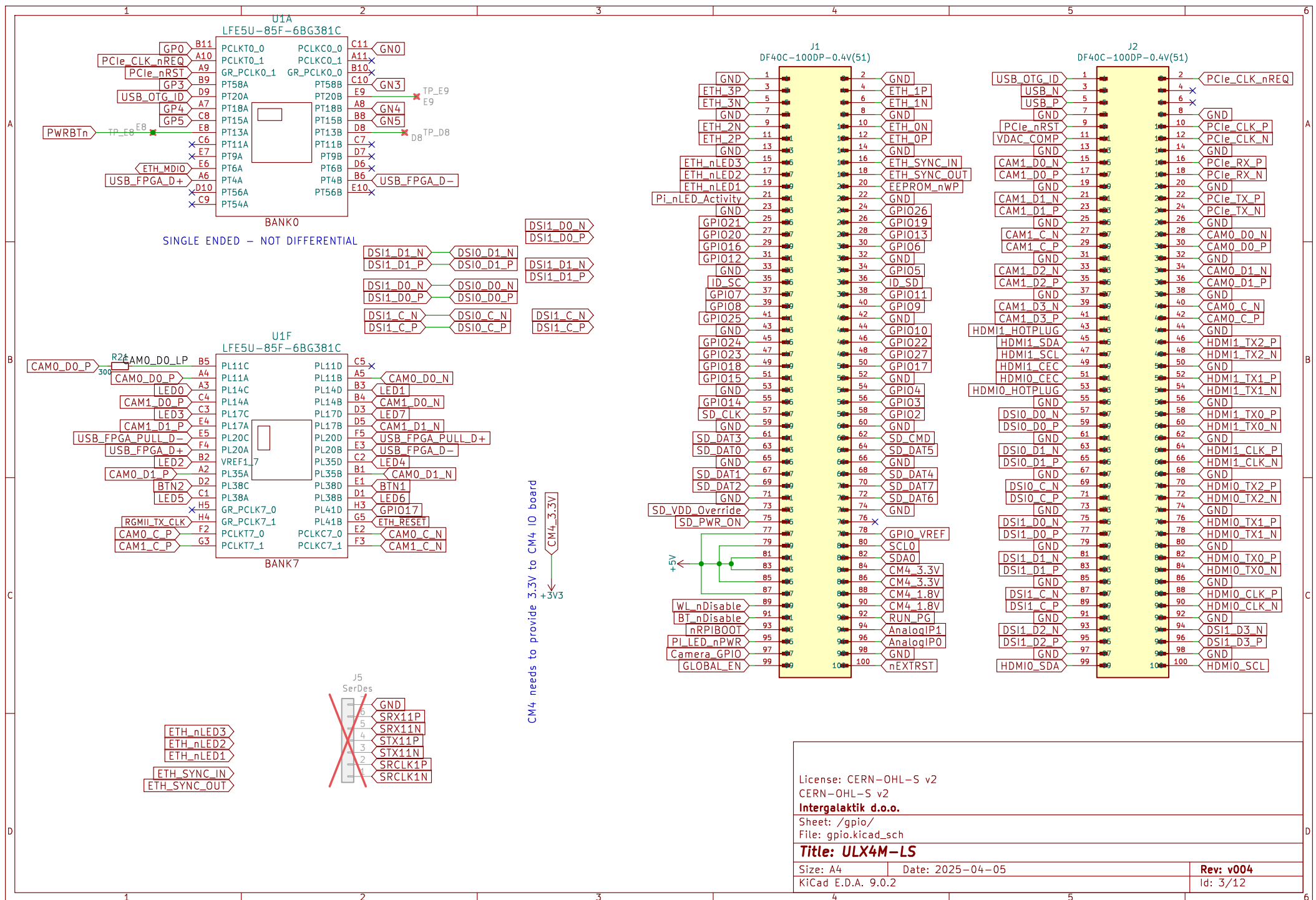
CERN-OHL-S v2
Intergalaktik d.o.o.

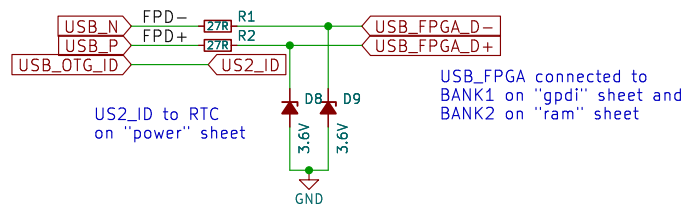
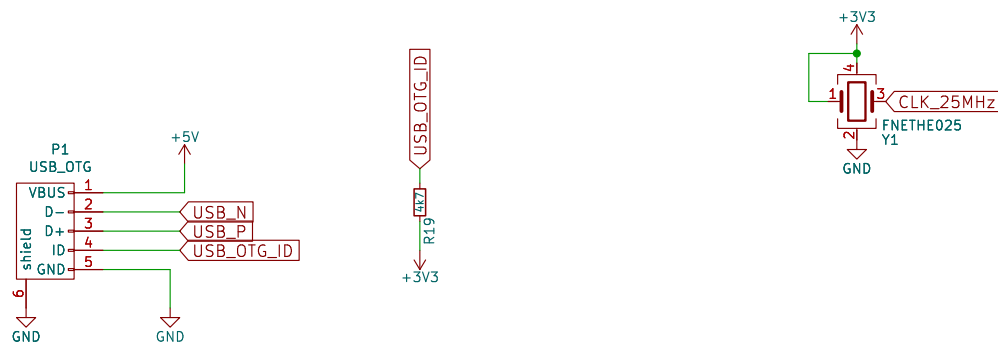
Sheet: /
File: ulx4m.kicad_sch

Title: ULX4M-LS

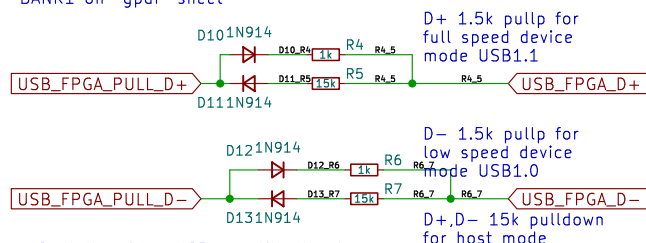
Size: A4 Date: 2025-04-05
KiCad E.D.A. 9.0.2

Rev: v004
Id: 1/12

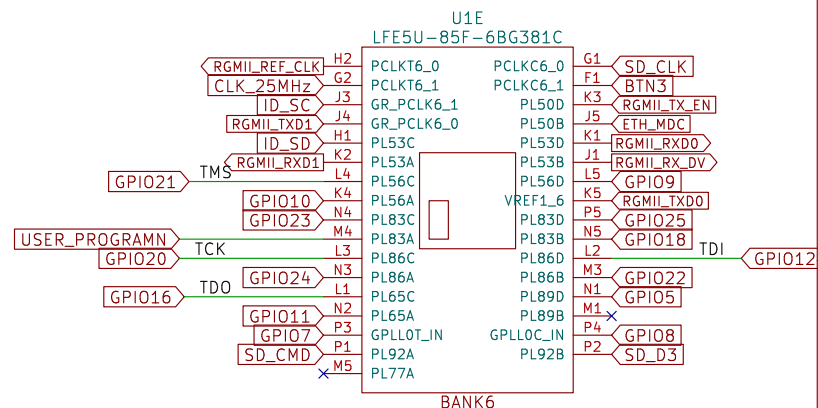
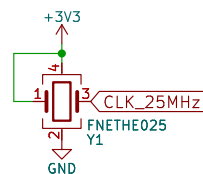




USB pull lines connected to BANK1 on "gpdj" sheet



Deviation from USB specification in pulldowns for BOM simplification. With series diode, correct value R63 R64 should be 12k but 15k is used.



FTDI default
CBUS0=TXDEN
CBUS1=nRXLED
CBUS2=nTXLED
CBUS3=nSLEEP



License: CERN-OHL-S v2
CERN-OHL-S v2
Intergalaktik d.o.o.

Sheet: /usb/
File: usb.kicad_sch

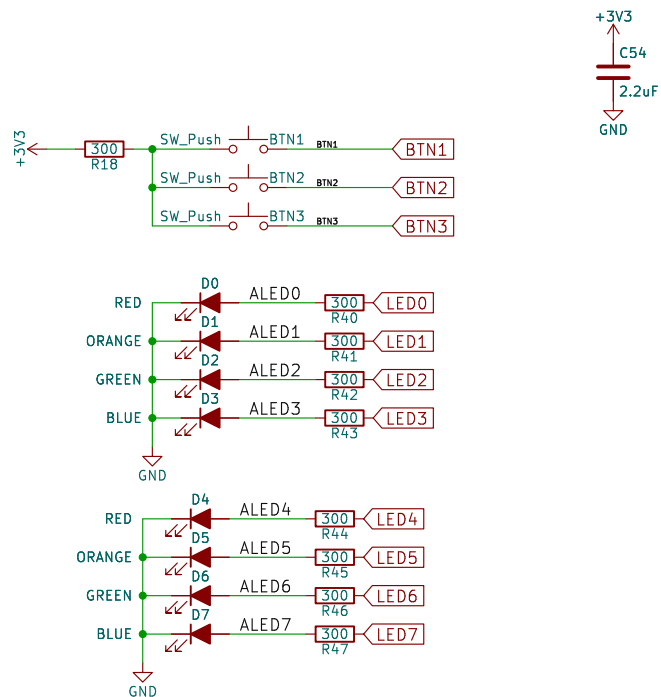
Title: ULX4M-LS

Size: A4 Date: 2025-04-05

KiCad E.D.A. 9.0.2

Rev: v004

Id: 4/12



License: CERN-OHL-S v2
CERN-OHL-S v2
Intergalaktik d.o.o.

Sheet: /blinky/
File: blinky.kicad_sch

Title: ULX4M-LS

Size: A4
KiCad E.D.A. 9.0.2

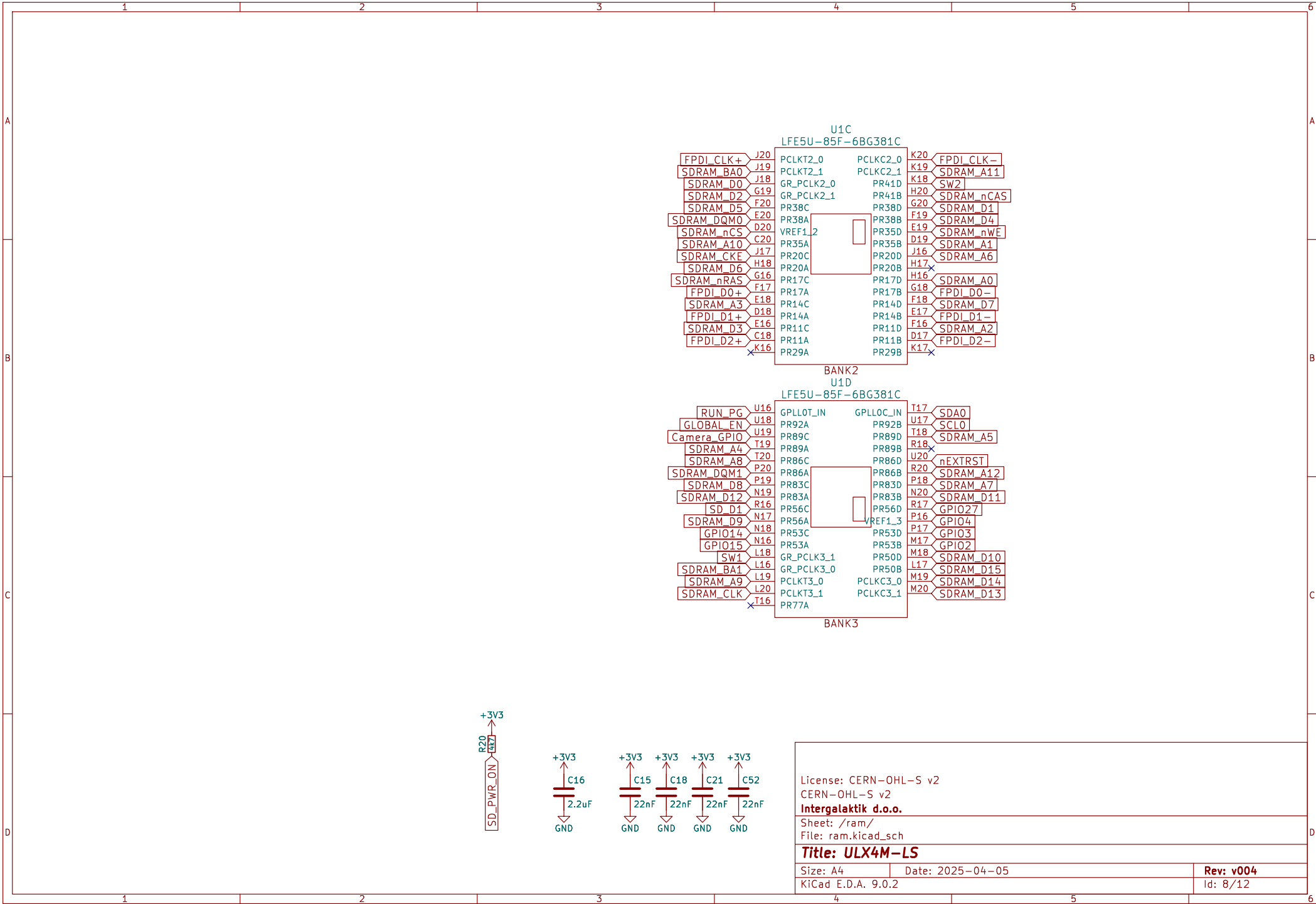
Date: 2025-04-05

Rev: v004
Id: 6/12



License: CERN-OHL-S v2
CERN-OHL-S v2
Intergalaktik d.o.o.
Sheet: /ethernet/
File: ethernet.kicad_sch
Title: ULX4M-LS

Size: A3	Date: 2025-04-05	Rev: v004
KiCad E.D.A. 9.0.2		Id: 7/12



License: CERN-OHL-S v2
CERN-OHL-S v2
Intergalaktik d.o.o.

Sheet: /ram/
File: ram.kicad_sch

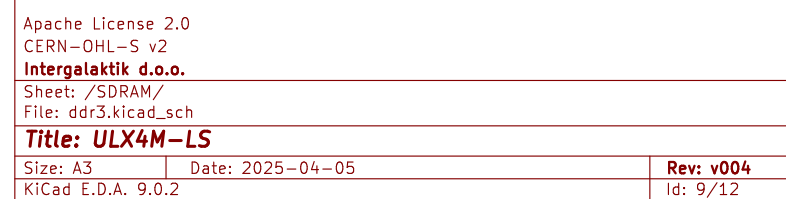
Title: ULX4M-LS

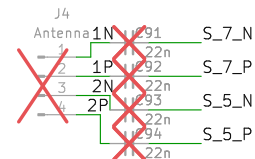
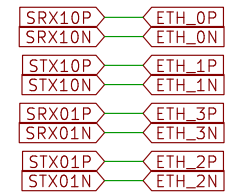
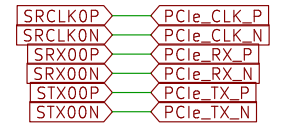
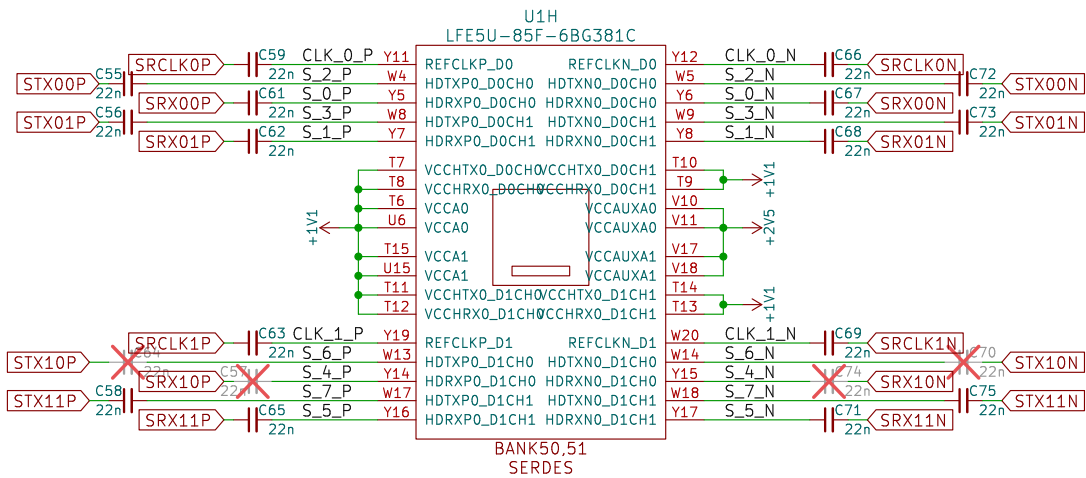
Size: A4 Date: 2025-04-05

KiCad E.D.A. 9.0.2

Rev: v004

Id: 8/12





Place shorts or caps for external antenna experiments

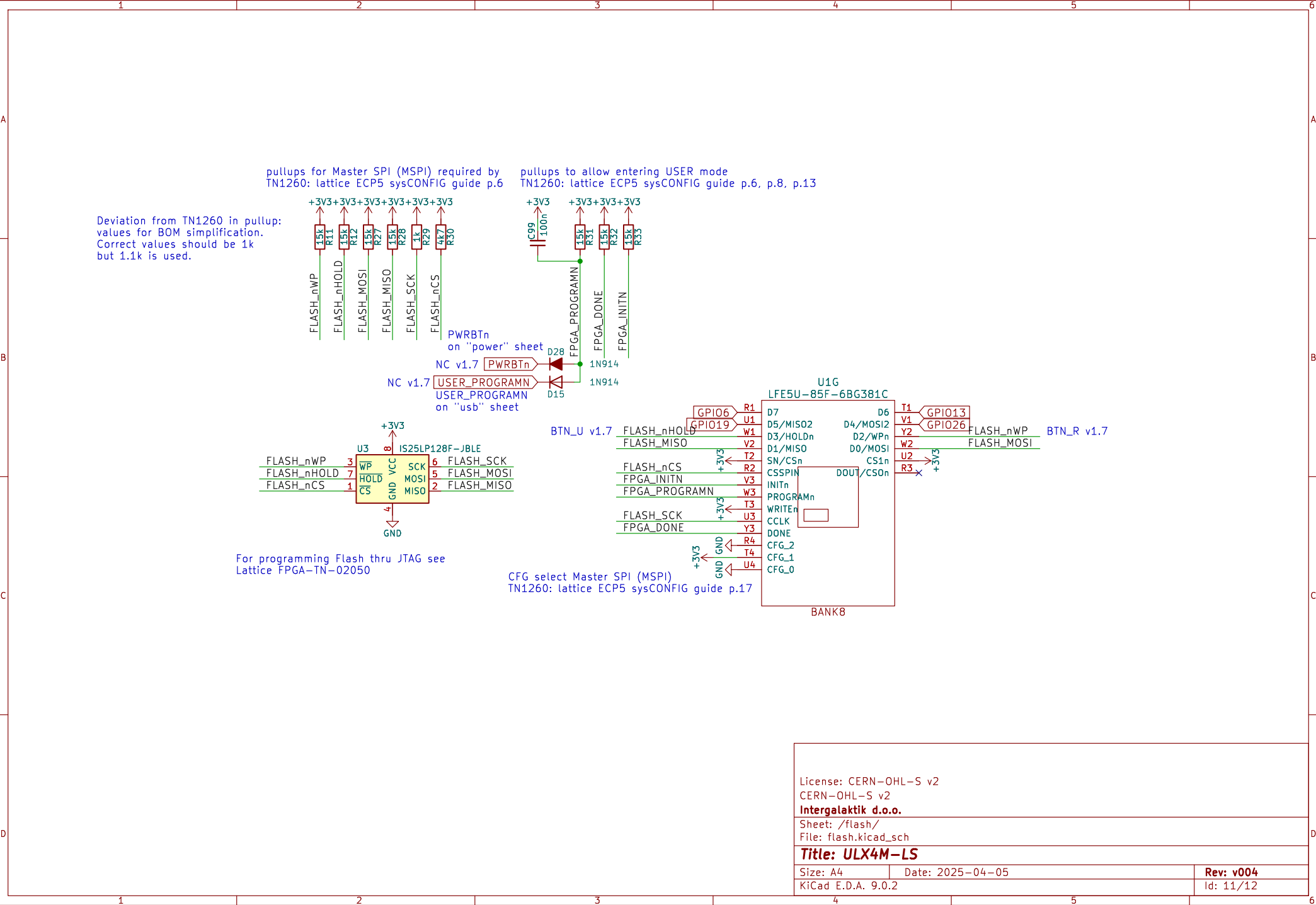
License: CERN-OHL-S v2
CERN-OHL-S v2
Intergalaktik d.o.o.

Sheet: /serdes/
File: serdes.kicad_sch

Title: ULX4M-LS

Size: A4 Date: 2025-04-05
KiCad E.D.A. 9.0.2

Rev: v004
Id: 10/12



License: CERN-OHL-S v2

CERN-OHL-S v2

Intergalaktik d.o.o.

Sheet: /flash/

File: flash.kicad_sch

Title: ULX4M-LS

Size: A4

Date: 2025-04-05

Rev: v004

KiCad E.D.A. 9.0.2

Id: 11/12

1	2	3	4	5	6
A					A
B					B
C					C
D					D
1	2	3	4	5	6

License: CERN-OHL-S v2
CERN-OHL-S v2

Intergalaktik d.o.o.

Sheet: /MechanicalDrawings/
File: mechanical.kicad_sch

Title: ULX4M-LS

Size: A4 Date: 2025-04-05

KiCad E.D.A. 9.0.2

Rev: v004

Id: 12/12